LECTURE 2
LINEAR REGRESSION MODEL AND OLS

Definitions

A common objective in econometrics is to study the effect of one group of variables, X_i, usually called the *regressors*, on another variable, Y_i, called the *dependent variable*. An econometrician observes the random data:

$$(Y_1, X_1), (Y_2, X_2), \ldots, (Y_n, X_n), \tag{1}$$

where for $i = 1, \ldots, n$, Y_i is a random variable and X_i is a random k-vector:

$$X_i = \begin{pmatrix} X_{i1} \\ X_{i2} \\ \vdots \\ X_{ik} \end{pmatrix}.$$

A pair (Y_i, X_i) is called an *observation*, and the collection of observations in (1) is called the *sample*. The vector X_i collects the values of k variables for observation i.

The joint distribution of (1) is called the *population*. The population does not correspond to any physical population, but to a probability space. In a *cross-sectional* framework (each observation is a different individual or a firm, etc.), it is often natural to assume that all observations are independently drawn from the same distribution. In this case, the population is described by the distribution of a single observation (Y_1, X_1), which is equivalent to saying that (Y_i, X_i) are iid for $i = 1, \ldots, n$. The iid assumption does not imply that Y_i and X_i are independent, but rather that the random vector (Y_i, X_i) is independent of (Y_j, X_j) for $i \neq j$. At the same time, Y_i and X_i can still be related.

In cross-sections, the relationship between the regressors and the dependent variable is modelled through the conditional expectation $\mathrm{E}[Y_i \mid X_i]$. The deviation of Y_i from its conditional expectation is called the *error* (or *disturbance*):

$$U_i = Y_i - \mathrm{E}[Y_i \mid X_i]. \tag{2}$$

Unlike X_i and Y_i, the error U_i is not observable, since the conditional expectation function is unknown to the econometrician.

In the *parametric* framework, it is assumed that the conditional expectation function depends on a number of unknown constants or *parameters*, and that the functional form of $\mathrm{E}[Y_i \mid X_i]$ is known. In the linear regression model, it is assumed that $\mathrm{E}[Y_i \mid X_i]$ is *linear in the parameters*:

$$\begin{aligned} \mathrm{E}[Y_i \mid X_i] &= \beta_1 X_{i1} + \beta_2 X_{i2} + \ldots + \beta_k X_{ik} \\ &= X_i^\top \beta, \end{aligned} \tag{3}$$

where

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{pmatrix}$$

is a k-vector of unknown constants. The linearity of $\mathrm{E}[Y_i \mid X_i]$ can be justified, for example, by saying that (Y_i, X_i) jointly has a multivariate normal distribution. Since $\beta_j = \frac{\partial \mathrm{E}[Y_i|X_i]}{\partial X_{ij}}$, the vector β is a vector of *marginal effects* of X_i, i.e., β_j gives the change in the conditional mean of Y_i per unit change in X_{ij}, while holding the values of other variables (X_{il} for $l \neq j$) fixed. One of the objectives is *estimation* of unknown β from the sample (1).

Combining equations (2) and (3), one can write:

$$Y_i = X_i^\top \beta + U_i. \tag{4}$$

By definition (2),

$$\mathrm{E}\left[U_i \mid X_i\right] = 0.$$

This implies that the regressors contain no information on the deviation of Y_i from its conditional expectation. Further, the Law of Iterated Expectation (LIE) implies that the errors have zero mean: $\mathrm{E}[U_i] = 0$. If (Y_i, X_i) are iid, then the errors $\{U_i : i = 1, \ldots, n\}$ are iid as well.

In the *classical regression model*, it is assumed that the variance of the errors U_i is independent of the regressors and the same for all observations:

$$\mathrm{Var}\left(U_i \mid X_i\right) = \sigma^2,$$

for some constant $\sigma^2 > 0$. This property is called *homoskedasticity*.

Assumptions

In this section, we formally define the linear regression model. We define

$$\begin{aligned} X &= \begin{pmatrix} X_1^\top \\ X_2^\top \\ \vdots \\ X_n^\top \end{pmatrix} \\ &= \begin{pmatrix} X_{11} & X_{12} & \ldots & X_{1k} \\ X_{21} & X_{22} & \ldots & X_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ X_{n1} & X_{n2} & \ldots & X_{nk} \end{pmatrix}, \\ Y &= \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \text{ and} \\ U &= \begin{pmatrix} U_1 \\ U_2 \\ \vdots \\ U_n \end{pmatrix}. \end{aligned}$$

The following are the four classical regression assumptions:

(A1) $Y = X\beta + U$.

(A2) $\mathrm{E}\left[U \mid X\right] = 0$ a.s.

(A3) $\mathrm{Var}\left(U \mid X\right) = \sigma^2 I_n$ a.s.

(A4) $\mathrm{rank}(X) = k$ a.s.[1]

[1] The column (row) rank of a matrix is the maximum number of linearly independent columns (rows). One can show that, for any matrix, the column and row ranks are equal. If A is an $n \times k$ matrix, then $\mathrm{rank}\left(A\right) \leq \min\left(n, k\right)$. If $\mathrm{rank}\left(A\right) = n$ (or $\mathrm{rank}\left(A\right) = k$), we say that A has full row (column) rank. Properties:

$$\begin{aligned} \mathrm{rank}\left(A\right) &= \mathrm{rank}\left(A^\top\right) = \mathrm{rank}\left(A^\top A\right) = \mathrm{rank}\left(AA^\top\right), \\ \mathrm{rank}\left(AB\right) &\leq \min\left(\mathrm{rank}\left(A\right), \mathrm{rank}\left(B\right)\right), \\ \mathrm{rank}\left(AB\right) &= \mathrm{rank}\left(A\right) \text{ if } B \text{ is square and of full rank.} \end{aligned}$$

Instead of conditioning on the observed values of the regressors, in the classical regression model, one can assume that X is not random, i.e., the values of X are fixed in repeated samples. In this case, (A2) becomes $\mathrm{E}[U]=0$ and (A3) becomes $\mathrm{Var}(U)=\sigma^2 I_n$. Since conditioning on X is basically equivalent to treating it as fixed, the two sets of assumptions lead to the same results.

For inference purposes, sometimes it is assumed that:

(A5) $U \mid X \sim N\left(0, \sigma^2 I_n\right)$.

In the case of fixed regressors, it is assumed instead that the unconditional distribution of the errors is normal. Assumptions (A1)-(A5) define the *classical normal regression model.* In this case,

$$Y \mid X \sim N\left(X\beta, \sigma^2 I_n\right).$$

Since all off-diagonal covariance elements in (A5) are zero, (A5) implies independence of the errors. Assumptions (A1)-(A4) alone do not imply independence across observations. Actually, for many important results, independence across observations is not required. Nevertheless, sometimes we would like to assume independence without normality:

(A6) $\{(Y_i, X_i) : i = 1, \ldots, n\}$ are iid.

In the case of fixed regressors, (A6) can be replaced with the assumption that $\{U_i : i = 1, \ldots, n\}$ are iid.

Assumption (A2) says that U is mean independent of X, which is a strong assumption. As mentioned above, it is equivalent to treating X as non-random, and in many economic applications it is hard to justify. However, many important results may be obtained with a weaker assumption of uncorrelatedness:

(A2*) For $i = 1, \ldots, n$, $\mathrm{E}\left[U_i X_i\right] = 0$ and $\mathrm{E}\left[U_i\right] = 0$.

Under (A2*), the expression $X_i^\top \beta$ does not have the interpretation of the conditional expectation. In this case, one can treat (4) as a *data-generating process*.

Assumption (A3) implies that the errors U_i have the same variance for all i and are uncorrelated with each other: $\mathrm{E}\left[U_i U_j \mid X\right] = 0$ for $i \neq j$. If desired, independence of the errors may be achieved through (A5) or Assumptions (A1) and (A6).

Assumption (A4) says that the columns of X are not linearly dependent. If it is violated, then one or more regressors simply duplicate the information contained in other regressors, and thus should be omitted.

Often, the first column of the matrix X is a column of ones. In this case, its coefficient β_1 is called the intercept. The intercept gives the average value of the dependent variable when all regressors are equal to zero.

Estimation by the method of moments

One of the objectives of econometric analysis is *estimation* of unknown parameters β and σ^2. An *estimator* is *any function* of the sample $\{(Y_i, X_i) : i = 1, \ldots, n\}$. An estimator cannot depend on the unobserved errors U_i or unknown parameters such as β directly; it can depend on them only through the observed variables Y and X. An estimator is generally not unique: several alternative estimators may exist for the same parameter.

One of the oldest methods of finding estimators is called the *method of moments* (MM). The MM says to replace population moments (expectations) with the corresponding sample moments (averages). Assumptions (A2) or (A2*) imply that at the true value of β,

$$\begin{aligned} 0 &= \mathrm{E}\left[U_i X_i\right] \\ &= \mathrm{E}\left[\left(Y_i - X_i^\top \beta\right) X_i\right]. \end{aligned} \tag{5}$$

Let $\widehat{\beta}$ be an estimator of β. According to the MM, we replace the expectation in (5) with the sample average:

$$\begin{aligned} 0 &= n^{-1} \sum_{i=1}^{n} \left(Y_i - X_i^\top \widehat{\beta}\right) X_i \\ &= n^{-1} \sum_{i=1}^{n} X_i Y_i - n^{-1} \sum_{i=1}^{n} X_i X_i^\top \widehat{\beta}. \end{aligned}$$

(Here, $Y_i - X_i^\top \widehat{\beta}$ is a scalar.) Solving for $\widehat{\beta}$, one obtains:

$$\begin{aligned}
\widehat{\beta} &= \left(n^{-1} \sum_{i=1}^{n} X_i X_i^\top \right)^{-1} n^{-1} \sum_{i=1}^{n} X_i Y_i \\
&= \left(\sum_{i=1}^{n} X_i X_i^\top \right)^{-1} \sum_{i=1}^{n} X_i Y_i \\
&= \left(X^\top X \right)^{-1} X^\top Y. \qquad (6)
\end{aligned}$$

The matrix $\sum_{i=1}^{n} X_i X_i^\top = X^\top X$ is invertible due to Assumption (A4). To show that $X^\top X = \sum_{i=1}^{n} X_i X_i^\top$, note that

$$\begin{aligned}
X^\top X &= \begin{pmatrix} X_1^\top \\ X_2^\top \\ \vdots \\ X_n^\top \end{pmatrix}^\top \begin{pmatrix} X_1^\top \\ X_2^\top \\ \vdots \\ X_n^\top \end{pmatrix} \\
&= \begin{pmatrix} X_1 & X_2 & \ldots & X_n \end{pmatrix} \begin{pmatrix} X_1^\top \\ X_2^\top \\ \vdots \\ X_n^\top \end{pmatrix} \\
&= X_1 X_1^\top + X_2 X_2^\top + \ldots + X_n X_n^\top \\
&= \sum_{i=1}^{n} X_i X_i^\top .
\end{aligned}$$

The expression $X_i^\top \widehat{\beta}$ gives the *estimated regression line*, with $\widehat{Y}_i = X_i^\top \widehat{\beta}$ being the *predicted* value of Y_i, and $\widehat{U}_i = Y_i - X_i^\top \widehat{\beta}$ being the *sample residual*,

$$\widehat{U} = Y - X\widehat{\beta}.$$

The vector $\widehat{U}$ is a function of the estimator of β. In the case of the MM estimator, the sample residuals have to satisfy the sample *normal equation*:

$$\begin{aligned}
0 &= X^\top \widehat{U} \qquad (7) \\
&= \sum_{i=1}^{n} \widehat{U}_i X_i \\
&= \begin{pmatrix} \sum_{i=1}^{n} \widehat{U}_i X_{i1} \\ \sum_{i=1}^{n} \widehat{U}_i X_{i2} \\ \vdots \\ \sum_{i=1}^{n} \widehat{U}_i X_{ik} \end{pmatrix}.
\end{aligned}$$

If the model contains an intercept, i.e., $X_{i1} = 1$ for all i, then the normal equation implies that $\sum_{i=1}^{n} \widehat{U}_i = 0$.

To estimate σ^2, write:

$$\begin{aligned}
\sigma^2 &= \mathrm{E}[U_i^2] \\
&= \mathrm{E}\left[\left(Y_i - X_i^\top \beta \right)^2 \right].
\end{aligned}$$

Since β is unknown, we must replace it by its MM estimator:

$$\widehat{\sigma}^2 = n^{-1} \sum_{i=1}^{n} \left(Y_i - X_i^\top \widehat{\beta} \right)^2 .$$

Least Squares

Let b be a candidate value for β. The *ordinary least squares* (OLS) estimator of β minimizes the *sum of squared errors*:

$$\begin{aligned} S(b) &= \sum_{i=1}^{n} \left(Y_i - X_i^\top b\right)^2 \\ &= (Y - Xb)^\top (Y - Xb). \end{aligned}$$

It turns out that $\widehat{\beta}$, the MM estimator presented in the previous section, is the OLS estimator as well. In order to show that, write

$$\begin{aligned} S(b) &= (Y - Xb)^\top (Y - Xb) \\ &= \left(Y - X\widehat{\beta} + X\widehat{\beta} - Xb\right)^\top \left(Y - X\widehat{\beta} + X\widehat{\beta} - Xb\right) \\ &= \left(Y - X\widehat{\beta}\right)^\top \left(Y - X\widehat{\beta}\right) \\ &\quad + \left(X\widehat{\beta} - Xb\right)^\top \left(X\widehat{\beta} - Xb\right) \\ &\quad +2\left(Y - X\widehat{\beta}\right)^\top \left(X\widehat{\beta} - Xb\right) \\ &= \left(Y - X\widehat{\beta}\right)^\top \left(Y - X\widehat{\beta}\right) \\ &\quad + \left(\widehat{\beta} - b\right)^\top X^\top X \left(\widehat{\beta} - b\right) \\ &\quad +2\widehat{U}^\top X \left(\widehat{\beta} - b\right) \text{ (equals zero by the normal equation)} \\ &= \left(Y - X\widehat{\beta}\right)^\top \left(Y - X\widehat{\beta}\right) + \left(\widehat{\beta} - b\right)^\top X^\top X \left(\widehat{\beta} - b\right). \end{aligned}$$

Minimization of $S(b)$ is equivalent to minimization of $\left(\widehat{\beta} - b\right)^\top X^\top X \left(\widehat{\beta} - b\right)$, because $\left(Y - X\widehat{\beta}\right)^\top \left(Y - X\widehat{\beta}\right)$ is not a function of b. If X is of full column rank, as assumed in (A4), $X^\top X$ is a positive definite matrix, and therefore

$$\left(\widehat{\beta} - b\right)^\top X^\top X \left(\widehat{\beta} - b\right) \geq 0,$$

where $\left(\widehat{\beta} - b\right)^\top X^\top X \left(\widehat{\beta} - b\right) = 0$ if and only if $\widehat{\beta} - b = 0$.

Alternatively, one can show that $\widehat{\beta}$ as defined in (6) is the OLS estimator, by taking the derivative of $S(b)$ with respect to b, and solving the first-order condition $\frac{dS(\widehat{\beta})}{db} = 0$. Write

$$S(b) = Y^\top Y - 2b^\top X^\top Y + b^\top X^\top Xb.$$

Using the fact that for a symmetric matrix A we have that

$$\frac{\partial x^\top Ax}{\partial x} = 2Ax,$$

the first-order condition is

$$\frac{\partial S\left(\widehat{\beta}\right)}{\partial b} = -2X^\top Y + 2X^\top X\widehat{\beta} = 0. \tag{8}$$

Solving for $\widehat{\beta}$, one obtains:

$$\widehat{\beta} = \left(X^\top X\right)^{-1} X^\top Y. \tag{9}$$

Note also that the first-order condition (8) can be written as $X^\top \left(Y - X\widehat{\beta}\right) = 0$, which gives us the normal equation (7).

Properties of $\widehat{\beta}$

1. $\widehat{\beta}$ is a *linear* estimator. An estimator b is linear if it can be written as $b = AY$, where A is some matrix that depends on X alone and does not depend on Y. In the case of the OLS, $A = \left(X^\top X\right)^{-1} X^\top$.

2. Under Assumptions (A1), (A2) and (A4), $\widehat{\beta}$ is an *unbiased* estimator, i.e.,

$$\mathrm{E}[\widehat{\beta}] = \beta.$$

In order to show unbiasedness, first, plug in the expression for Y in Assumption (A1) into equation (9):

$$\begin{aligned} \widehat{\beta} &= \left(X^\top X\right)^{-1} X^\top \left(X\beta + U\right) \\ &= \beta + \left(X^\top X\right)^{-1} X^\top U. \end{aligned} \tag{10}$$

Next, consider the conditional expectation of $\widehat{\beta}$ given X:

$$\begin{aligned} \mathrm{E}\left[\widehat{\beta} \mid X\right] &= \mathrm{E}\left[\beta + \left(X^\top X\right)^{-1} X^\top U \mid X\right] \\ &= \beta + \mathrm{E}\left[\left(X^\top X\right)^{-1} X^\top U \mid X\right]. \end{aligned}$$

Now,

$$\begin{aligned} \mathrm{E}\left[\left(X^\top X\right)^{-1} X^\top U \mid X\right] &= \left(X^\top X\right)^{-1} X^\top \mathrm{E}\left[U \mid X\right] \\ &= 0, \end{aligned}$$

since, by Assumption (A2), $\mathrm{E}\left[U \mid X\right] = 0$. Therefore,

$$\mathrm{E}\left[\widehat{\beta} \mid X\right] = \beta. \tag{11}$$

Finally, the LIE implies that

$$\begin{aligned} \mathrm{E}[\widehat{\beta}] &= \mathrm{E}[\mathrm{E}[\widehat{\beta} \mid X]] \\ &= \beta. \end{aligned}$$

Equation (11) shows that $\widehat{\beta}$ is conditionally unbiased given X. Unbiasedness cannot be established under Assumption (A2*) alone.

3. Under Assumptions (A1), (A2) and (A4),

$$\mathrm{Var}\left(\widehat{\beta} \mid X\right) = \left(X^\top X\right)^{-1} X^\top \mathrm{E}\left[UU^\top \mid X\right] X \left(X^\top X\right)^{-1}.$$

Under homoskedastic errors, i.e., Assumption (A3), the expression for the conditional variance simplifies to

$$\mathrm{Var}\left(\widehat{\beta} \mid X\right) = \sigma^2 \left(X^\top X\right)^{-1}.$$

To show this, we start with the definition of the variance:

$$
\begin{aligned}
\operatorname{Var}\left(\widehat{\beta} \mid X\right) &= \mathrm{E}\left[\left(\widehat{\beta}-\mathrm{E}\left[\widehat{\beta} \mid X\right]\right)\left(\widehat{\beta}-\mathrm{E}\left[\widehat{\beta} \mid X\right]\right)^{\top} \mid X\right] \\
&= \mathrm{E}\left[\left(\widehat{\beta}-\beta\right)\left(\widehat{\beta}-\beta\right)^{\top} \mid X\right] \\
&= \mathrm{E}\left[\left(X^{\top} X\right)^{-1} X^{\top} U U^{\top} X\left(X^{\top} X\right)^{-1} \mid X\right] \text{ (follows from (10))} \\
&= \left(X^{\top} X\right)^{-1} X^{\top} \mathrm{E}\left[U U^{\top} \mid X\right] X\left(X^{\top} X\right)^{-1}.
\end{aligned}
$$

Under homoskedasticity, $\mathrm{E}\left[U U^{\top} \mid X\right]=\sigma^2 I_n$, and

$$
\begin{aligned}
\left(X^{\top} X\right)^{-1} X^{\top} \mathrm{E}\left[U U^{\top} \mid X\right] X\left(X^{\top} X\right)^{-1} &= \left(X^{\top} X\right)^{-1} X^{\top}\left(\sigma^2 I_n\right) X\left(X^{\top} X\right)^{-1} \\
&= \sigma^2\left(X^{\top} X\right)^{-1} X^{\top} X\left(X^{\top} X\right)^{-1} \\
&= \sigma^2\left(X^{\top} X\right)^{-1}.
\end{aligned}
$$

In the case of fixed regressors, $\operatorname{Var}\left(\widehat{\beta}\right)=\sigma^2\left(X^{\top} X\right)^{-1}$.

4. If we add normality of the errors, i.e., under Assumptions (A1)-(A5), we obtain the following result:

$$
\widehat{\beta} \mid X \sim N\left(\beta, \sigma^2\left(X^{\top} X\right)^{-1}\right).
$$

It is sufficient to show that, conditional on X, the distribution of $\widehat{\beta}$ is normal. Then, $\widehat{\beta} \mid X \sim N\left(\mathrm{E}\left[\widehat{\beta} \mid X\right], \operatorname{Var}\left(\widehat{\beta} \mid X\right)\right)$. However, normality of $\widehat{\beta} \mid X$ follows from the facts that $\widehat{\beta}$ is a linear function of Y, and that $Y \mid X$ is normal by Assumption (A5) (see the normal distribution section in Lecture 1). Again, in the case of fixed regressors, we simply omit conditioning on X,

$$
\widehat{\beta} \sim N\left(\beta, \sigma^2\left(X^{\top} X\right)^{-1}\right).
$$

5. *Efficiency* or the Gauss-Markov Theorem: Under Assumptions (A1)-(A4), the OLS estimator is the Best Linear Unbiased Estimator of β (BLUE), where "best" means having the smallest variance: for any linear and unbiased estimator b, we have that $\operatorname{Var}(b \mid X)-\operatorname{Var}\left(\widehat{\beta} \mid X\right)$ is a positive semi-definite matrix:

$$
\operatorname{Var}(b \mid X)-\operatorname{Var}\left(\widehat{\beta} \mid X\right) \geq 0.
$$

Furthermore, if $\tilde{\beta}$ is a linear unbiased estimator and $\operatorname{Var}(\tilde{\beta} \mid X)=\operatorname{Var}\left(\widehat{\beta} \mid X\right)$, then $\tilde{\beta}=\widehat{\beta}$ with probability one.
Note that since the theorem discusses the conditional variance of the OLS estimator, unbiasedness in the statement of the theorem actually refers to unbiasedness conditional on X, i.e., $\mathrm{E}[b \mid X]=\beta$.

Proof: Let b be a linear and unbiased estimator of β :

$$
\begin{aligned}
b &= AY, \\
\mathrm{E}[b \mid X] &= \beta.
\end{aligned}
$$

These conditions imply that $AX=I_k$ with probability 1. Indeed,

$$
\begin{aligned}
\mathrm{E}[b \mid X] &= \mathrm{E}[A(X\beta+U) \mid X] \\
&= AX\beta+A\,\mathrm{E}[U \mid X].
\end{aligned}
$$

By Assumption (A2), $\mathrm{E}[U \mid X] = 0$, and thus, in order to obtain unbiasedness, we need $AX = I_k$. Next, we show that $\mathrm{Cov}\left(\widehat{\beta}, b \mid X\right) = \mathrm{Var}\left(\widehat{\beta} \mid X\right)$:

$$\begin{aligned}
\mathrm{Cov}\left(\widehat{\beta}, b \mid X\right) &= \mathrm{E}\left[\left(\widehat{\beta} - \beta\right)(b - \beta)^\top \mid X\right] \\
&= \mathrm{E}\left[\left(X^\top X\right)^{-1} X^\top U U^\top A^\top \mid X\right] \\
&= \left(X^\top X\right)^{-1} X^\top \mathrm{E}\left[U U^\top \mid X\right] A^\top \\
&= \sigma^2 \left(X^\top X\right)^{-1} X^\top A^\top \text{ (since, by Assumption (A3), } \mathrm{E}\left[U U^\top \mid X\right] = \sigma^2 I_n) \\
&= \sigma^2 \left(X^\top X\right)^{-1} \text{ (since } X^\top A^\top = I_k) \\
&= \mathrm{Var}\left(\widehat{\beta} \mid X\right).
\end{aligned}$$

Finally,

$$\begin{aligned}
\mathrm{Var}\left(\widehat{\beta} - b \mid X\right) &= \mathrm{Var}\left(\widehat{\beta} \mid X\right) - \mathrm{Cov}\left(\widehat{\beta}, b \mid X\right) - \mathrm{Cov}\left(b, \widehat{\beta} \mid X\right) + \mathrm{Var}\left(b \mid X\right) \\
&= \mathrm{Var}\left(b \mid X\right) - \mathrm{Var}\left(\widehat{\beta} \mid X\right). \qquad (12)
\end{aligned}$$

Now, since any variance-covariance matrix is positive semi-definite, we have that

$$\mathrm{Var}\left(b \mid X\right) - \mathrm{Var}\left(\widehat{\beta} \mid X\right) \geq 0.$$

To show uniqueness, suppose that there is a linear unbiased estimator $\tilde{\beta}$ such that $\mathrm{Var}(\tilde{\beta} \mid X) = \mathrm{Var}\left(\widehat{\beta} \mid X\right)$. Then, by (12), $\mathrm{Var}\left(\widehat{\beta} - \tilde{\beta} \mid X\right) = 0$, and therefore $\tilde{\beta} = \widehat{\beta} + c(X)$ for some k-vector-valued function $c(X)$ that can depend only on X. However, since both $\widehat{\beta}$ and $\tilde{\beta}$ are conditionally unbiased given X, it follows that $c(X) = 0$ with probability one. □

Note that Assumption (A3), $\mathrm{E}\left[U U^\top \mid X\right] = \sigma^2 I_n$, plays a crucial role in the proof of the result. We could not draw conclusions about the efficiency of OLS without it.